

AVL Trees (Balancing)

Self-balancing BST with rotations.

Key concepts

- Balance factor = $\text{height}(\text{left}) - \text{height}(\text{right})$, kept in $[-1,1]$.
- Rotations: LL, RR, LR, RL.
- Keeps operations $O(\log n)$.

Key idea / example

An insert that unbalances triggers a rotation to rebalance.

Today's tasks (1.5h)

- ☐ Learn the 4 rotations
- ☐ Insert with rebalancing
- ☐ Explain balance factor

Resources

Playlist: youtube.com/playlist?list=PLsr8vTgyLdy_YndxNcI4WkH5Vorj5qvrV

(Clickable playlist link is in the app: Goals > this day > Study block.)